

VEL'MO, Russian Federation

WMO: 239820

Lat: 61.0168N

Lon: 93.4281E

Elev: 122

StdP: 99.87

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.6	-43.9	-49.8	0.0	-47.4	-46.1	0.0	-43.8	7.1	-18.9	6.4	-21.4	0.2	130	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	29.2	19.0	27.2	18.5	25.1	17.7	20.7	26.7	19.6	25.3	18.3	23.6	2.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.7	23.7	17.4	12.6	22.1	16.2	11.7	21.1	60.7	26.6	56.5	25.6	52.5	23.4	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	6.0	5.2	DB	-48.9	31.6	4.1	1.9	-51.8	33.0	-54.2	34.1	-56.5	35.2	-59.4	36.6	(4)
(5)			WB	-49.4	22.9	4.8	1.3	-52.9	23.9	-55.7	24.6	-58.4	25.4	-62.0	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-27.3	-21.7	-10.4	-2.0	4.8	13.8	17.6	13.7	6.5	-3.0	-17.3	-21.8		
	DBStd	16.50	9.77	8.91	7.90	5.10	4.33	5.07	3.31	3.62	4.26	5.94	9.27	10.75		
	HDD10.0	5572	1156	888	634	362	171	23	0	9	121	404	819	984		
	HDD18.3	8140	1414	1121	892	612	418	149	54	150	355	662	1069	1243		
	CDD10.0	524	0	0	0	0	11	137	237	123	16	0	0	0		
	CDD18.3	51	0	0	0	0	0	13	32	5	0	0	0	0		
	CDH23.3	847	0	0	0	0	9	330	428	76	4	0	0	0		
	CDH26.7	218	0	0	0	0	2	94	112	11	0	0	0	0		
	WSAvg	1.8	1.3	1.4	2.1	2.6	2.4	2.0	1.9	1.5	1.8	1.8	1.6	1.4		
	PrecAvg	576	31	22	25	31	48	63	50	79	76	57	54	37		
	PrecMax	688	67	48	60	80	105	116	117	143	141	116	87	82		
	PrecMin	446	13	1	2	12	17	13	12	12	36	23	21	15		
	PrecStd	73	13	11	13	16	19	26	22	39	27	21	15	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.7	-1.1	7.4	13.1	23.7	31.7	31.6	28.1	23.3	11.8	1.0	-2.1		
		MCWB	-6.5	-2.2	2.4	6.9	13.3	20.5	19.4	19.0	16.7	6.8	-0.2	-2.7		
	2%	DB	-8.4	-3.4	4.4	10.1	20.1	29.4	29.5	25.2	18.8	7.9	-0.8	-3.7		
		MCWB	-9.1	-4.4	0.4	4.4	11.8	18.8	19.2	18.2	13.6	5.1	-1.5	-4.4		
	5%	DB	-11.1	-5.8	2.3	7.7	16.5	27.2	27.7	23.3	15.7	6.0	-2.5	-5.4		
		MCWB	-11.6	-6.8	-0.2	3.0	9.4	17.9	19.3	17.2	11.8	4.1	-3.2	-6.0		
	10%	DB	-13.4	-8.7	0.6	5.5	13.0	24.9	25.6	21.3	13.4	4.2	-3.8	-7.2		
		MCWB	-13.8	-9.4	-1.3	1.9	7.9	17.1	18.2	16.1	10.4	2.8	-4.4	-7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.4	-2.3	2.9	7.0	14.3	22.1	22.3	21.2	17.4	7.7	0.1	-2.8		
		MCDB	-5.8	-1.1	6.7	12.2	22.3	28.5	27.7	25.4	22.1	10.3	0.6	-2.4		
	2%	WB	-9.2	-4.3	1.0	5.2	12.5	20.2	21.1	19.2	14.6	5.6	-1.6	-4.3		
		MCDB	-8.5	-3.5	3.7	8.9	18.5	28.0	26.8	23.8	17.4	7.6	-1.0	-3.7		
	5%	WB	-11.6	-6.7	-0.2	3.5	10.5	18.8	20.0	17.9	12.3	4.3	-3.1	-5.9		
		MCDB	-11.2	-5.9	1.9	6.8	15.9	25.4	25.7	22.4	15.0	5.7	-2.6	-5.3		
	10%	WB	-13.9	-9.4	-1.4	2.2	8.2	17.3	19.0	16.6	10.6	2.7	-4.5	-7.8		
		MCDB	-13.4	-8.7	0.6	5.0	12.4	23.7	24.2	20.1	13.1	4.2	-3.9	-7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	12.4	13.7	11.0	11.1	13.6	13.1	11.6	8.7	6.6	9.4	9.1		
		MCDBR	10.5	10.0	12.3	13.7	17.8	18.9	16.7	15.9	14.1	8.6	7.5	6.9		
		MCWBR	9.9	9.2	9.3	8.8	10.1	9.7	8.3	9.4	8.8	6.0	7.2	6.6		
	5% WB	MCDBR	10.5	9.7	11.1	11.7	16.0	17.1	14.3	13.4	12.4	7.9	7.1	7.0		
		MCWBR	10.0	8.9	8.5	8.1	9.8	9.7	8.2	8.5	8.3	5.9	7.0	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.297	0.328	0.359	0.374	0.389	0.439	0.410	0.348	0.314	0.253	0.228		
	taud		1.940	1.992	2.012	2.050	2.262	2.377	2.220	2.276	2.421	2.260	1.996	1.884		
	Ebn at Noon		420	644	765	821	846	839	777	770	767	644	429	285		
	Edh at Noon		46	83	115	135	119	108	124	108	78	64	42	29		
(44) All-Sky Solar Radiation	RadAvg		0.39	1.29	2.63	4.08	4.89	5.51	4.96	3.55	2.12	0.93	0.44	0.19		
	RadStd		0.05	0.10	0.28	0.43	0.44	0.62	0.45	0.29	0.33	0.15	0.07	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	+3.08	+2.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page